The Problem Ladder: Mathematics of Every Target

A reference for the OURS problem bank

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This walks down the tier list and, for each problem, gives the object, its closed form, the derivation or proof, and the boxed **What the machine fits** note that names the exact quantity to regress. The tiers are ordered by how hard the *fitting* is; within a tier the entries are independent. This is reference material, written for the advisor and the students who will prove these by hand.

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1 Tier 1: one-shot wins

1.1 Circle area, $A = \pi r^2$

The area is defined by $g(r) = \int_0^r 4\sqrt{r^2 - x^2}\, dx$. Substituting $x = r\sin\theta$,

$$g(r) = 4\int_0^{\pi/2} r^2 \cos^2\theta\, d\theta = 4r^2 \cdot \frac{\pi}{4} = \pi r^2,$$

using $\int_0^{\pi/2} \cos^2\theta\, d\theta = \pi/4$.

> Fit a polynomial $\sum_j c_j r^j$ to the numerically integrated $g(r)$. The discovery is the single nonzero coefficient $c_2 = \pi$.

1.2 Arcsine and logarithm from integrals

Arcsine. From $\arcsin r = \int_0^r (1-x^2)^{-1/2}\, dx$ and the binomial series $(1-x^2)^{-1/2} = \sum_{n\geq 0} \frac{(2n)!}{4^n (n!)^2} x^{2n}$, integrate term by term:

$$\arcsin r = \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n+1)}\, r^{2n+1} = r + \frac{r^3}{6} + \frac{3r^5}{40} + \cdots.$$

Logarithm. From $\ln(1+x) = \int_0^x \frac{dt}{1+t}$ and $\frac{1}{1+t} = \sum_{n\geq 0} (-1)^n t^n$,

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}\, x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots.$$

> Fit a low-degree polynomial to the integral data and recover the leading coefficients; prove the general term by the series above plus induction. Lesson: the answer is a series, so the machine recovers a few terms and the human supplies the pattern.

1.3 Power-law areas, $I_\alpha(r) = C(\alpha)\, r^{2\alpha+1}$

For $\alpha > -1$, $I_\alpha(r) = \int_0^r (r^2 - x^2)^\alpha\, dx$. With $x = r\sin\theta$,

$$I_\alpha(r) = r^{2\alpha+1} \int_0^{\pi/2} \cos^{2\alpha+1}\theta\, d\theta = C(\alpha)\, r^{2\alpha+1}, \qquad C(\alpha) = \frac{\sqrt{\pi}}{2}\, \frac{\Gamma(\alpha+1)}{\Gamma(\alpha+\frac{3}{2})}.$$

A check: $\alpha = \frac{1}{2}$ gives the circle quadrant, $C(\frac{1}{2}) = \pi/4$ and exponent 2, so $4I_{1/2}(r) = \pi r^2$.

> Fit $\log I_\alpha(r) = \log C(\alpha) + (2\alpha+1)\log r$. The discovery is twofold: the exponent $2\alpha+1$ and the Gamma-function constant $C(\alpha)$.

2 Tier 2: spot the pattern

2.1 Exponential from $x' = ax$

Separating $x' = ax$, $x(0) = 1$ gives $x = e^{at} = \sum_{n\geq 0} \frac{(at)^n}{n!}$. With the bivariate hypothesis $\tilde{x}(a,t) = \sum_{m,n} A_{mn} a^m t^n$, matching the Maclaurin series forces

$$A_{mn} = \frac{\delta_{mn}}{n!}, \qquad \text{i.e.} \qquad \tilde{x}(a,t) = \sum_n \frac{a^n t^n}{n!}.$$

> Fit the coefficient table A_{mn}. The discovery is the diagonal-plus-factorial pattern $A_{mn} = \delta_{mn}/n!$.

2.2 Logistic growth, $x' = rx(1-x)$

With $x(0) = x_0$ the closed form is

$$x(t) = \frac{x_0}{x_0 + (1-x_0)e^{-rt}}.$$

For learning, the short-time expansion is the target. Differentiating the ODE repeatedly, $x' = rx - rx^2$, $x'' = rx'(1-2x)$, and so on, gives Taylor coefficients

$$x(t) = x_0 + rx_0(1-x_0)\,t + \tfrac{1}{2}r^2 x_0(1-x_0)(1-2x_0)\,t^2 + \cdots.$$

> Fit a low-order series in t (for fixed x_0, r, or as a polynomial in all three). Unlike the exponential, the coefficients have no clean closed form, so the pattern is harder to read; recover them and verify against the ODE.

2.3 Monomial moments, $\mu_{2k} = \frac{2}{2k+1}$

For the uniform weight on $[-1,1]$,

$$\mu_k = \int_{-1}^{1} x^k\,dx = \begin{cases} \dfrac{2}{k+1}, & k \text{ even}, \\ 0, & k \text{ odd}, \end{cases} \qquad \text{so} \qquad \mu_{2k} = \frac{2}{2k+1}.$$

> Compute the moments numerically and fit μ_{2k} as a function of k. The discovery is the rational rule $\mu_{2k} = 2/(2k+1)$ and the vanishing of odd moments. Good warm-up for the Legendre problem below.

2.4 Catalan numbers from the semicircle

The semicircle density on $[-R,R]$ is $\rho_R(x) = \frac{2}{\pi R^2}\sqrt{R^2 - x^2}$ (normalized, since the half-disk has area $\pi R^2/2$). Its even moments, via $x = R\sin\theta$, are

$$m_{2k} = \frac{4R^{2k}}{\pi}\int_0^{\pi/2} \sin^{2k}\theta\cos^2\theta\,d\theta = \frac{4R^{2k}}{\pi}\cdot\frac{1}{2}B\big(k+\tfrac{1}{2},\tfrac{3}{2}\big).$$

Using $\Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$, $\Gamma(k+\frac{1}{2}) = \frac{(2k)!}{4^k k!}\sqrt{\pi}$, $\Gamma(k+2) = (k+1)!$,

$$m_{2k} = \frac{R^{2k}(2k)!}{4^k\, k!\,(k+1)!} = \frac{C_k R^{2k}}{4^k}, \qquad C_k = \frac{1}{k+1}\binom{2k}{k}.$$

> Compute m_{2k}, divide by $R^{2k}/4^k$, and recover the integers $1, 1, 2, 5, 14, 42, \ldots$. Validate with $C_{k+1} = \frac{2(2k+1)}{k+2} C_k$. The moments of the semicircle law being Catalan numbers is the doorway to free probability and random matrix theory.

2.5 The Legendre three-term recurrence

The polynomials orthogonal on $[-1, 1]$ (weight 1) satisfy, in monic form, $p_{k+1} = (x - a_k)p_k - b_k p_{k-1}$. **Why only three terms:** expand $xp_k = \sum_{j=0}^{k+1} c_j p_j$ and pair with p_i, giving $c_i \|p_i\|^2 = \langle xp_k, p_i\rangle = \langle p_k, xp_i\rangle$; since xp_i has degree $i+1$ and $p_k \perp$ everything of degree $< k$, this vanishes for $i < k-1$, leaving only $i = k-1, k, k+1$. The coefficients are

$$a_k = \frac{\langle xp_k, p_k\rangle}{\langle p_k, p_k\rangle} = 0 \text{ (odd integrand)}, \qquad b_k = \frac{\langle p_k, p_k\rangle}{\langle p_{k-1}, p_{k-1}\rangle} = \frac{k^2}{4k^2 - 1}.$$

Check: $b_1 = \frac{1}{3}$ gives $p_2 = x^2 - \frac{1}{3}$, the monic Legendre polynomial.

> Estimate moments μ_j, build the inner products, and fit (a_k, b_k) versus k. The discovery is $a_k = 0$ and $b_k = k^2/(4k^2 - 1)$: a recurrence, not a closed form.

3 Tier 3: now there is a square root

3.1 The quadratic formula

For $x^2 + bx + c = 0$, complete the square: $(x + \frac{b}{2})^2 = \frac{b^2}{4} - c$, so

$$x = \frac{-b \pm \sqrt{b^2 - 4c}}{2}.$$

The two roots satisfy Vieta's relations $r_+ + r_- = -b$ and $r_+ r_- = c$, and the radical is the root difference, $r_+ - r_- = \sqrt{b^2 - 4c}$. The discriminant $b^2 - 4c$ decides real (> 0) versus complex (< 0) roots.

> Fit $\tilde{u} = \xi_0 + \xi_1 b + \xi_2 c + \xi_3\sqrt{\xi_4 + \cdots + \xi_7 b^2 + \xi_9 bc}$, nonlinear in the parameters, so the loss is nonconvex and the two roots are two basins. Note the scaling degeneracy: $\xi_3\sqrt{\text{inside}}$ is unchanged under $\xi_3 \to \xi_3/\lambda$, inside $\to \lambda^2 \cdot$inside, so only combinations such as $\xi_6/\xi_7 = -4$ are identifiable. Normalize $\xi_7 = 1$ to pin the rest. The easy half is symmetric ($r_+ + r_-$); the hard half is the radical.

4 Tier 4: nested radicals

4.1 The cubic (Cardano)

Depress $x^3 + ax^2 + bx + c = 0$ by $x = t - a/3$ to get $t^3 + pt + q = 0$, with $p = b - \frac{a^2}{3}$, $q = c - \frac{ab}{3} + \frac{2a^3}{27}$. Set $t = u + v$; then $t^3 - 3uv\,t - (u^3 + v^3) = 0$ matches the depressed cubic when $uv = -\frac{p}{3}$ and

$u^3 + v^3 = -q$. So u^3, v^3 solve the resolvent $z^2 + qz - \frac{p^3}{27} = 0$, giving

$$t = \sqrt[3]{-\tfrac{q}{2} + \sqrt{\tfrac{q^2}{4} + \tfrac{p^3}{27}}} + \sqrt[3]{-\tfrac{q}{2} - \sqrt{\tfrac{q^2}{4} + \tfrac{p^3}{27}}}.$$

With $\omega = e^{2\pi i/3}$ and one consistent pair (u_0, v_0) obeying $u_0 v_0 = -p/3$, the roots are $u_0 + v_0$, $\omega u_0 + \omega^2 v_0$, $\omega^2 u_0 + \omega v_0$. The discriminant is $\Delta = -4p^3 - 27q^2$, and $\frac{q^2}{4} + \frac{p^3}{27} = -\frac{\Delta}{108}$. When $\Delta > 0$ there are three real roots but the inner square root is imaginary: this is the **casus irreducibilis**, where the real roots provably cannot be written in real radicals. The real-arithmetic alternative is trigonometric,

$$t_k = 2\sqrt{-\tfrac{p}{3}}\cos\Big[\tfrac{1}{3}\arccos\big(\tfrac{3q}{2p}\sqrt{-3/p}\big) - \tfrac{2\pi k}{3}\Big], \quad k = 0, 1, 2 \quad (\Delta \geq 0).$$

> Fit the real-radical hypothesis in (p, q) for the one-real-root regime, true coefficients $-\frac{1}{2}, 1, \frac{1}{4}, \frac{1}{27}$. Sweep the (p, q) plane: the fit fails exactly where $4p^3 + 27q^2 > 0$, so the machine rediscovers the casus irreducibilis as a region a real formula cannot reach. The boundary is the discriminant curve.

4.2 The quartic (Ferrari) and the quintic wall

Quartic. Depress to $t^4 + pt^2 + qt + r = 0$. Introduce a parameter m and complete a square:

$$\big(t^2 + \tfrac{p}{2} + m\big)^2 = 2m\, t^2 - q\, t + \big(\tfrac{p^2}{4} + pm + m^2 - r\big).$$

The right side is a perfect square in t exactly when its discriminant vanishes, which gives the **resolvent cubic**

$$8m^3 + 8p\, m^2 + (2p^2 - 8r)\, m - q^2 = 0.$$

Solve it for one m (a cubic, hence Tier 4 nesting), after which both sides are squares and the quartic factors into two quadratics. So the quartic reduces to a cubic that reduces to quadratics.

Quintic wall. The general quintic has *no* solution in radicals (Abel–Ruffini). The reason is Galois theory: a polynomial is solvable by radicals iff its Galois group is solvable, and the generic quintic has group S_5, which is not solvable because A_5 is simple and nonabelian.

> For the quartic, fit the resolvent-cubic structure and verify the factorization. For the quintic, the honest experiment is a *negative* result: no finite real-radical hypothesis in the coefficients can fit the general quintic's roots, and the fit error stays bounded away from zero no matter how many radical terms you add.

5 Tier 5: approximately equal

5.1 Stirling's approximation

From $\Gamma(n+1) = \int_0^\infty e^{\,n \ln x - x}\, dx$, the exponent $\phi(x) = n \ln x - x$ peaks at $x = n$ with $\phi(n) = n \ln n - n$ and $\phi''(n) = -1/n$. Laplace's method (Gaussian approximation at the peak) gives

$$n! = \Gamma(n+1) \sim e^{\phi(n)} \sqrt{\frac{2\pi}{|\phi''(n)|}} = \sqrt{2\pi n}\, \Big(\frac{n}{e}\Big)^n.$$

Equivalently, $\ln n! = \sum_{k \leq n} \ln k \approx \int_1^n \ln x\, dx + \frac{1}{2} \ln n + \cdots$ by Euler–Maclaurin.

Fit $\ln n!$ to the ansatz $n\ln n - n + \frac{1}{2}\ln n + C$ and recover $C = \frac{1}{2}\ln(2\pi)$. The discovery is the constant $\sqrt{2\pi}$ and the $\sqrt{n}$ prefactor.

5.2 The prime number theorem

$\pi(N) \sim \dfrac{N}{\ln N}$. Heuristically the density of primes near x is about $1/\ln x$, so

$$\pi(N) \approx \int_2^N \frac{dx}{\ln x} = \mathrm{Li}(N) \sim \frac{N}{\ln N}.$$

The full proof (Hadamard, de la Vallée Poussin) rests on the nonvanishing of $\zeta(s)$ on $\mathrm{Re}(s) = 1$ and is well beyond the warm-up, but the leading asymptotic is what a fit recovers.

Fit $\pi(N)$ against $N/\ln N$ and against $\mathrm{Li}(N)$ over a range of N. The discovery is the leading order, and the better fit of $\mathrm{Li}(N)$ over $N/\ln N$ is itself an observation worth making.

6 Tier 6: best possible constants

6.1 Wirtinger's inequality

For 2π-periodic f with $\int_0^{2\pi} f = 0$,

$$\int_0^{2\pi} f^2\,dx \le \int_0^{2\pi} (f')^2\,dx,$$

with equality iff $f = a\cos x + b\sin x$. Proof by Fourier series: write $f = \sum_{n\neq 0} c_n e^{inx}$ (no constant term). Then $\int f^2 = 2\pi\sum_{n\neq 0}|c_n|^2$ and $\int (f')^2 = 2\pi\sum_{n\neq 0} n^2|c_n|^2$, so

$$\int (f')^2 - \int f^2 = 2\pi \sum_{n\neq 0}(n^2-1)|c_n|^2 \ge 0,$$

since $n^2 \ge 1$, with equality iff only the $n = \pm 1$ modes are present. The sharp constant 1 is the smallest eigenvalue of $-d^2/dx^2$ on mean-zero periodic functions, so this is an eigenvalue problem in disguise.

Estimate the best constant by minimizing $\int (f')^2 / \int f^2$ over a basis, and recover both the constant 1 and the extremizer $\sin x$.

6.2 The isoperimetric inequality

For a simple closed plane curve with perimeter P and enclosed area A,

$$4\pi A \le P^2,$$

with equality iff the curve is a circle. It follows from Wirtinger (Hurwitz's proof). Parametrize by $t \in [0, 2\pi]$ proportional to arc length so $x'^2 + y'^2 = (P/2\pi)^2$, and translate so $\int x\,dt = 0$. With $A = \int_0^{2\pi} x\,y'\,dt$,

$$\frac{P^2}{2\pi} - 2A = \int_0^{2\pi} \big(x'^2 + y'^2 - 2xy'\big)dt = \underbrace{\int (x'^2 - x^2)\,dt}_{\ge 0 \text{ (Wirtinger)}} + \underbrace{\int (x - y')^2\,dt}_{\ge 0} \ge 0,$$

so $P^2 \geq 4\pi A$. Equality forces $x = a\cos t + b\sin t$ and $y' = x$, a circle.

Fit the best constant in $P^2 \geq c\,A$ over families of curves and recover $c = 4\pi$, with the circle as the extremizer. This is the capstone: the same inequality, Wirtinger, drives both Tier-6 problems.

7 The common thread

Across all six tiers the structure is identical: the machine returns numbers, and those numbers are a conjecture until proved. Tier 1 and 2 conjectures are closed by direct integration, series, and the Beta and Gamma evaluations. Tier 3 and 4 are closed by completing the square and by Cardano and Ferrari, with Galois theory marking the quintic wall. Tier 5 is closed by Laplace's method and Euler–Maclaurin. Tier 6 is closed by Fourier analysis and the eigenvalue reading of the sharp constant. Build each experiment to produce the numbers, then write the proof. The proof is the deliverable.